

A Data Channel over the Find My Network

Hundreds of millions of finder devices relay beacons for anyone. Prior work showed it carries data — **what kind of channel is it?**

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the full report

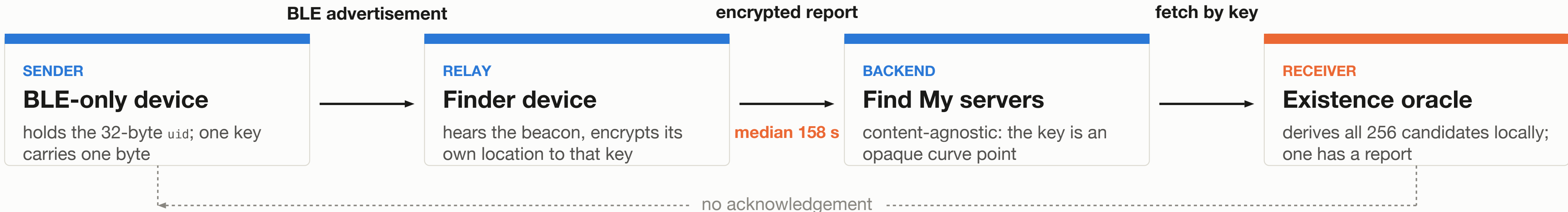
99.5–99.94 %
median deliverability at full transmit power

0
payload corruptions

158_s
median propagation latency

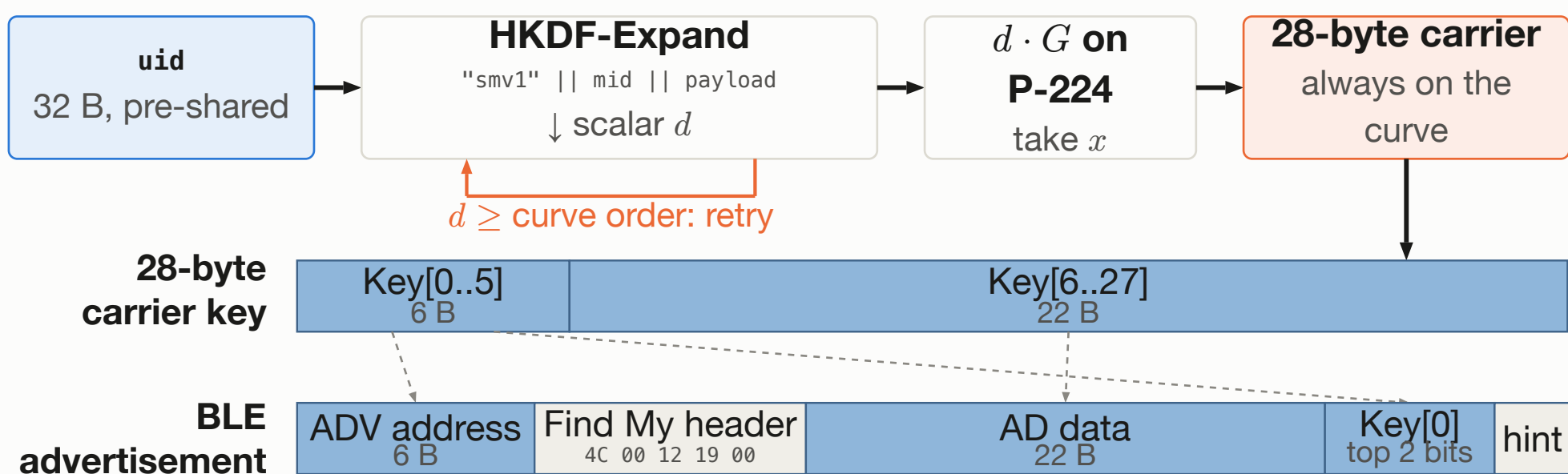
0.06–1.2 byte/s
goodput, depending on reliability trade-off

The channel, end to end — one-way, store-and-forward, unacknowledged



One byte, one curve point

- Carrier layer** — which key to advertise: HKDF-Expand under the `uid` → scalar d , carrier = $x(d \cdot G)$ on NIST P-224
- Link layer** — those 28 bytes scattered across the settable regions; to any scanner, an ordinary Find My advertisement
- Recovery is an existence oracle** — 256 possible carriers per `mid`; the one with a report on file names the byte

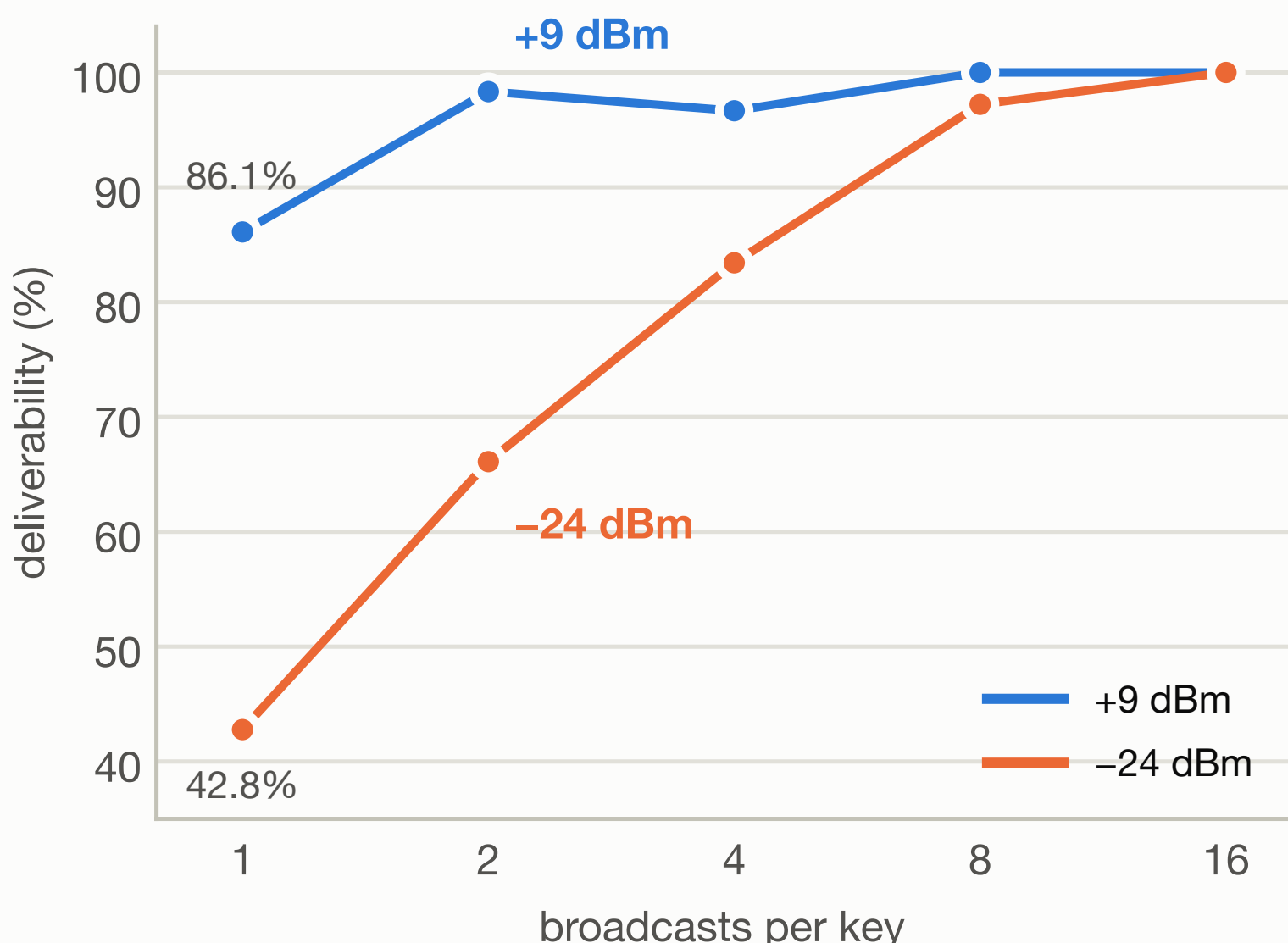


Every carrier valid by construction — a missing report is never an encoding fault, only network loss: a clean **erasure channel**

The uid makes it private end to end — otherwise the candidate space is the whole curve, and the reports stay encrypted

Repetition versus power

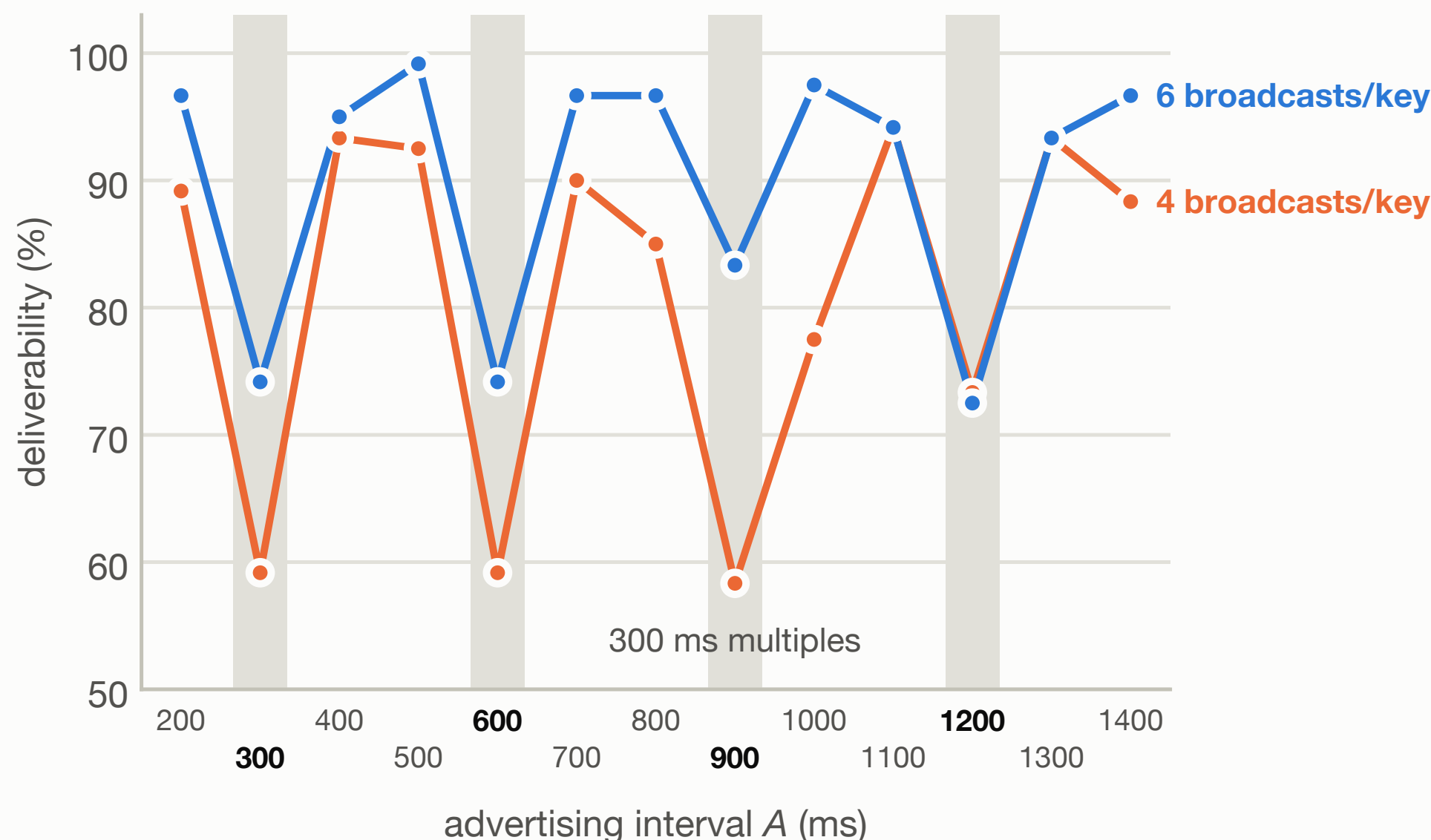
- Broadcasts per key B : the most effective control** — misses decay roughly geometrically, 42.8% to 100% as B goes 1 to 16
- Power sets not the shape but where the curve saturates** — independent Bernoulli chances to be heard
- At full power the rotation period is inert** — no headroom left



Deliverability against broadcasts per key, at -24 dBm and +9 dBm.

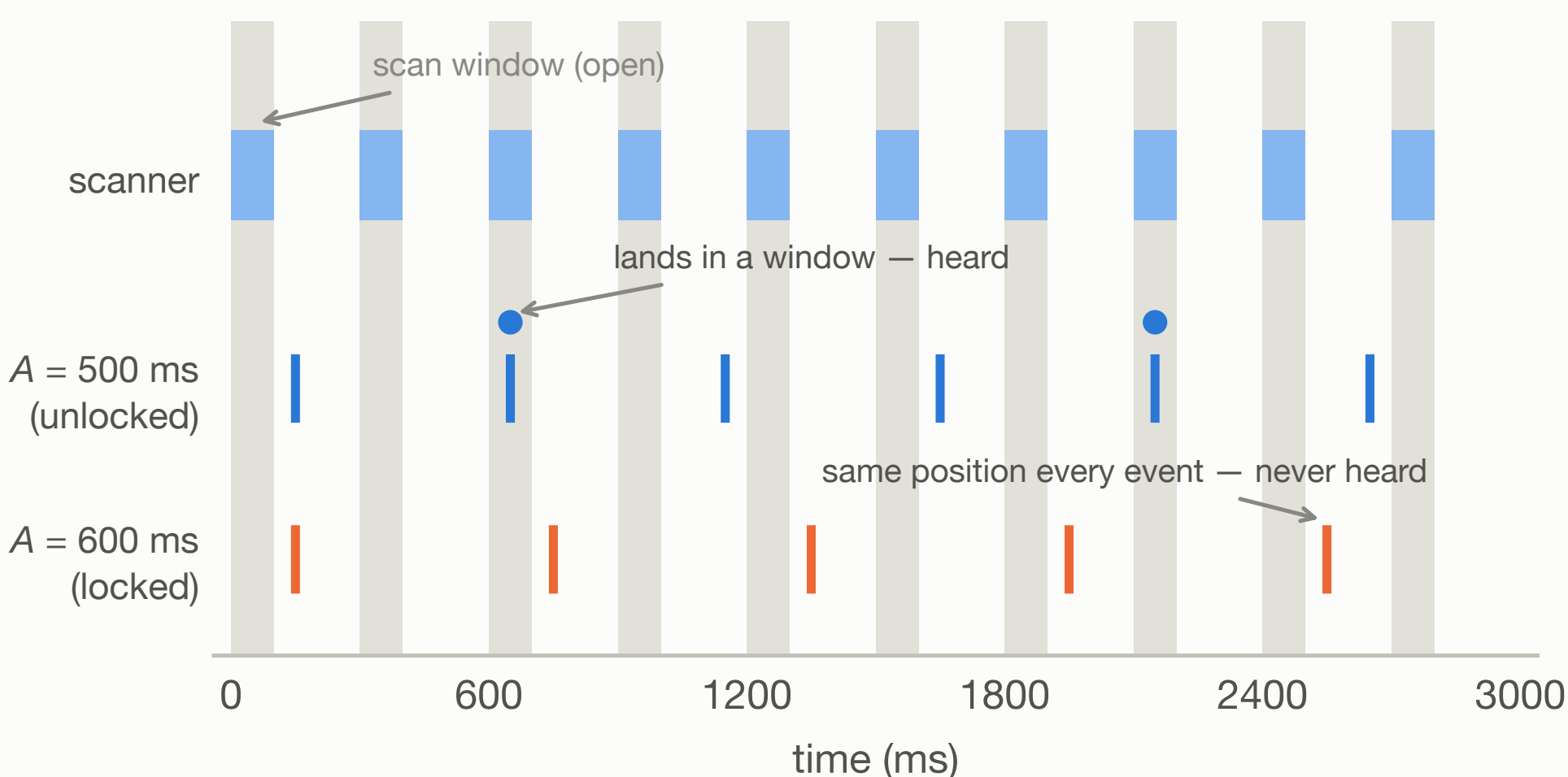
The 300 ms deliverability nulls

- Sharp, regularly spaced deficits at every multiple of 300 ms** — 69.3% against 92.7% at the surrounding intervals ($p < 5 \times 10^{-5}$)
- B pinned at 4 and 6: deficits at the *same* intervals in both arms → the penalty tracks A , not the rotation period



Deliverability against advertising interval, 200–1400 ms at -24 dBm, at 4 and 6 broadcasts per key.

- Plain arithmetic** — a scanner opens a window once per cycle, deaf for the rest; an interval that is a multiple of ~ 300 ms keeps landing at the same position, and in the deaf gap every broadcast misses alike
- Deliverability tracks the distinct positions reached, $300 / \gcd(A, 300)$ — 48.3% stuck at one, 85.0% at six



Scan-cycle geometry; window width and starting offset illustrative, not measured.

A natural hope: rescue a locked interval by broadcasting more. It cannot — at $B = 16$, 98.9% at $A = 500$ ms against 76.7% at 600 ms. Synchronised repeats are not independent opportunities.

What the channel is

- A pure erasure channel** — corrupts nothing, loses little at full power, never delivers quickly
- The delay is not in discovery** — store-and-forward, relay-bound latency
- Transmit rules** — at least eight broadcasts per key; never near a multiple of 300 ms
- Goodput-reliability trade-off** — varying with transmit power
- The nulls measure the unobserved scanners** — a coordinated fraction of finders on a shared periodic schedule
- Verdict** — suited to asynchronous, delay-tolerant messaging; unsuited for anything real-time